What is a System on a Chip?

A System on a chip is an IC (integrated circuit) that integrates the components of a computer and other electronic systems. It may contain various types of signal functions and is not limited by any one type of functional capability. The importance of SoCs are that they are compatible, cheap to manufacture and can handle processes with low power consumption. Due to this reason, they find their application in almost all modern day gadgets like phones, handhelds, media players, calculators etc. The idea is to build a circuit as per requirement. Assume the requirement to be that of making a functional timekeeping digital watch (As was the case of the first SoC in the 1970s). This function is carried out by combining a basic timekeeping circuit to an LCD screen and voila! You have a digital watch. The question that comes to mind is how is this different from a micro controller. An SoC integrates micro-controllers with advanced peripherals like GPUs, Wi-Fi etc. A good way to put it would be to say that an SoC is for micro controller what a micro controller is for processors. There is no one correct build or blueprint for an SoC. The build entirely depends on the objective that is expected to be achieved by the SoC. In certain cases, an SoC is not adequate to get the job done. In such cases SiPs (System in package) are used. We shall learn more about them later. Let us see what components an SoC comprises of.

Inside an SoC

A SoC does not have a fixed blueprint, however there are certain components that every SoC has.

1. A micro controller, microprocessor or a digital signal processor. This is the core of an SoC and provides the basic memory to connect with the other peripherals.
2. Memory Blocks which according to the requirement may be a selection from RAM, ROM or even flash memory. Again, it entirely depends on the task at hand for the device.
3. Oscillators and phase loops to keep time. Major breakthroughs have been made in this field and now we have circuits which are capable of keeping time using simple digital logic. This, in theory, would double the efficiency of SoCs.
4. External peripherals like USB, Ethernet, Fire Wire.
5. Power management circuits

A Bus is also an important component in an SoC and the selection of the